



**UNIVERSITY
OF LONDON**

MSc Data Science

DSM120 Financial Data Modelling

Midterm coursework assignment: April to September 2026 study session

The coursework is worth 100 marks, weighted at 50% of the final mark.

- Please note you are permitted to upload your coursework in the final submission area as many times as you like before the deadline. You will receive a similarity/originality score which represents what the Turnitin system identifies as work similar to another source. The originality score can take over 24 hours to generate, especially at busy times e.g. submission deadline.
- If you upload the wrong version of your coursework, you can resubmit updated versions of your coursework. You need to click on the 'submit paper' button again and submit your new version before the deadline.

In doing so, this will delete the previous version that you submitted, and your new updated version will replace it. Therefore, your Turnitin similarity score should not be affected. If there is a change in your Turnitin similarity score, it will be due to any changes you may have made to your coursework.

- Please note, when the due date is reached, the version you have submitted last, will be considered as your final submission and it will be the version that is marked.
- **Once the due date has passed, it will not be possible for you to upload a different version of your assessment. Therefore, you must ensure you have submitted the correct version of your assessment which you wish to be marked, by the due date.**

Background and Coursework Introduction

This coursework is designed to support the learning outcomes of the module by giving you hands-on experience in analysing real-world time-based data, with a particular focus on financial contexts. The aim is not simply to implement methods, but to develop the analytical judgement to select, apply, and critically evaluate them across a range of datasets.

The coursework accompanies the module progressively. As each topic is taught, you will apply the methods and ideas to a personal portfolio of time series data. That portfolio forms the basis of all three parts of the assessment, so the dataset choices you make at the start will carry through to the end.

The assessment is structured in three parts. Parts 1 and 2 follow directly from the taught topics, while Part 3 is more open-ended, reflecting the module's broader objective of equipping you with the ability to investigate, analyse, and make predictions on complex time-based data using both established and modern computational techniques.

Once your portfolio is established, the coursework tasks build on it topic by topic, from basic visualisation and stationarity analysis through to autoregressive modelling, Monte Carlo simulation, and open-ended exploration.

Dataset requirements

You must build a portfolio of at least six time series, including:

- at least two high-frequency financial datasets (minute-level data);
- a diverse selection of datasets, for example:
 - trending and non-trending series
 - seasonal and non-seasonal series
 - relatively stable and more volatile series.

You may select a subset of any high-frequency dataset; you are not required to use the full three-year range.

You are expected to:

- justify your dataset choices
- highlight differences in behaviour across datasets.

If the datasets used are available publicly, or accessible on the VLE, you do not need to upload them in your submission. Instead, please provide a direct link to the dataset within your code.

Submission requirements

Students are required to submit the following:

1. **A Jupyter notebook**

This should contain the relevant code and any additional supporting material. Please make sure the work is reproducible when the marking team runs it locally.

2. **A report**

Ideally, this should be an HTML file generated from the Jupyter Notebook, detailing the methodology, analysis, and findings.

The coursework assignment

Part 1

At the end of each of the first five topics, you will find a section outlining what you should include in your portfolio at that stage. Your submission should provide a structured account of your engagement with each topic, demonstrating both implementation and critical analysis. This should include:

- A summary of your approach, describing the steps you have taken and the reasoning behind your choices.
- Results presented clearly, using graphs, error estimates, or any other appropriate method to communicate findings effectively.
- The software and code used, ensuring reproducibility and clarity in implementation.
- A reflection on your findings, discussing observations about the datasets and methods explored.
- A critical evaluation of the models applied, assessing their limitations, and the insights gained from your analysis.

You are encouraged to use a diverse range of datasets across your portfolio, so that you can explore how different data characteristics, such as trend, seasonality, and volatility, influence modelling choices and forecasting performance.

[60 marks]

Part 2

In this section, you will extend your analysis by carrying out ACF, PACF, and AR(q) modelling at two different frequencies of a selected high-frequency dataset. For example, you may extract the data at intervals of 5 minutes and 10 minutes and use these to perform a comparative analysis.

Use Monte Carlo methods in conjunction with AR(q) to simulate 25 potential future scenarios for each frequency. Use kernel density estimation (KDE) to visualise the approximate distribution of the simulated future scenarios and to support discussion of forecast uncertainty.

You should compare and evaluate forecasting performance across both frequencies, considering how model behaviour and usefulness may vary with time resolution. You are encouraged to explore whether alternative forecasting approaches, parameter refinements, or additional comparisons can deepen your analysis.

This part is intended to assess not only technical implementation, but also your ability to interpret uncertainty and evaluate the practical usefulness of the models.

[20 marks]

Part 3

The final part of the coursework is more open-ended. You should extend your analysis in two ways: first, by applying techniques already covered in the module in a new or deeper way; and second, by exploring an approach that has not been covered in detail during the lectures. To achieve the highest marks, you should engage meaningfully with both aspects of this exploration.

For the first aspect, you may revisit any method from the taught content and push it further. For example, you could compare different approaches, vary parameters systematically, apply a method to a new type of dataset, or combine techniques in a way not demonstrated in the lectures.

For the second aspect, you have freedom to go beyond the taught content. Suitable choices might include a modern forecasting method, such as LSTMs, Transformers, Prophet, or gradient boosting models applied to time series; a domain-specific technique from quantitative finance, such as pairs trading and cointegration, technical analysis indicators, or regime-switching models; or a deeper extension of a method already introduced in the module.

The aim is not simply to use more advanced techniques, but to apply your chosen methods appropriately to your portfolio data and reflect on what they contribute to your analysis.

[20 marks]

How this coursework will be graded

The marks in this coursework are allocated as follows:

Part 1

There are 60 marks available for completing the tasks outlined at the end of each of the first five topics. These tasks should be implemented clearly and reproducibly, but you are not expected to build everything from scratch. It is perfectly acceptable to use *statsmodels* or other appropriate Python libraries. For each of the five topics:

- Up to 4 marks for producing the results of the main algorithms on selected data. Full marks will be awarded for well-structured implementation with appropriate visualisation.
- Up to 4 marks for reflecting on how well the method works under different conditions, with clear discussion of its strengths and limitations.
- Up to 4 marks for exploring beyond the standard implementation, such as comparing multiple approaches, experimenting with different parameters, or applying additional evaluation metrics.

Part 2

There are 20 marks available for completing this part:

- Up to 8 marks for conducting ACF, PACF, and AR(q) analysis at two different frequencies, demonstrating correct implementation and a well-reasoned discussion of the results.
- Up to 8 marks for applying Monte Carlo methods in conjunction with AR(q), ensuring well-structured simulations, appropriate visualisations, and insightful interpretation, including Kernel Density Estimation (KDE) to visualise the approximate distribution of future scenarios.
- Up to 4 marks for extending the analysis beyond standard implementation, such as comparing different forecasting approaches, evaluating performance variations across time resolutions, or refining parameter selection.

Part 3

The remaining 20 marks are more open-ended, allowing you to draw on your own interests, creativity, and background knowledge. There are several ways to obtain these marks, and it is up to you to put them together.

- Up to 10 marks for extending or deepening your use of methods from the taught content. This could involve applying a familiar technique in a new context, comparing approaches, or combining methods in a way not demonstrated in the lectures. You may combine more than one form of extension, but the total available for this aspect is 10 marks.

- Up to 10 marks for investigating an approach that was not covered in detail in the lectures. Your submission should include both implementation and an evaluation of its results against your portfolio data.

References

Alongside the course materials, for essential reading on time series forecasting and analysis, please refer to:

- Rob J. Hyndman & George Athanasopoulos (2021), *Forecasting: Principles and Practice* ([Online Book](#)) – A widely used, frequently updated resource covering key forecasting concepts.
- Franses, van Dijk, & Opschoor, *Time Series Models for Business and Economic Forecasting* – Focuses on the role of time series in economic modeling.
- James Douglas Hamilton (1994), *Time Series Analysis* – A comprehensive, mathematically rigorous reference for advanced time series methods.

A full list of reading materials is available on the VLE page.

[END OF COURSEWORK ASSIGNMENT]